EE4-27
EE9-CS5-2
EE9-FPN2-08

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

SYSTEMS IDENTIFICATION

Wednesday, 12 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Parisini
Second Marker(s) : F. Teng

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SYSTEMS IDENTIFICATION

Recommendations for candidates:

The exam has to be carried out remotely following the official indications provided by the College. Moreover, the algebraic calculations have to be carried out using a standard calculator. You can use the casiofx991es (or equivalent), the calculator on your phone, a simple calculator online or on your PC, such as the examples here. You are not allowed to use a graphic calculator, MATLAB, Mathematica, Maple, R, Python, Wolfram Alpha, etc. We are relying on your academic integrity in following this requirement.

For all these aspects, please check the Assessment FAQ at:

https://www.imperial.ac.uk/electrical-engineering/study/current-students-course-handbook/covid19/assessment-faq/

1. Consider an object that is moving along a one-dimensional trajectory. Its position over time $p(t)$ is described by the stochastic model:

$$p(t+1) = \frac{2}{3} \cdot p(t) - \frac{1}{3} \cdot p(t-1) + \eta(t)$$

where $\eta(\cdot) \sim WN\left(0, \frac{1}{3}\right)$ is a white stochastic process.

Two position sensors provide the measurements $s_1(t)$ and $s_2(t)$ of the variable $p(t)$ according to the measurement equations:

$$s_1(t) = \frac{5}{4} \cdot p(t) + \frac{1}{6} \cdot e_1(t), \qquad s_2(t) = \frac{1}{2} \cdot p(t) + e_2(t),$$

where $e_1(\cdot) \sim WN(1,1)$ is a non-zero mean white stochastic process (that is, it is a white stochastic process to which the constant 1 has been added) and $e_2(\cdot) \sim WN(0,1)$ is a white stochastic process. The stochastic processes $e_1(\cdot)$, $e_2(\cdot)$, and $\eta(\cdot)$ are supposed to be mutually independent.

a) Show that the stochastic process $p(\cdot)$ is stationary.

[4 Marks]

b) Compute the expected value $\mathbb{E}[p(\cdot)]$ of the stochastic process $p(\cdot)$.

[3 Marks]

c) Compute the first two samples $\gamma_p(0), \gamma_p(1)$ of the correlation function of the stochastic process $p(\cdot)$.

[6 Marks]

d) Determine the optimal values of the parameters α, β and γ of the minimum variance unbiased linear estimator

$$\hat{p}(t) = \alpha \cdot s_1(t) + \beta \cdot s_2(t) + \gamma$$

yielding an estimate $\hat{p}(t)$ of the object position $p(t)$ on the basis of the measurements $s_1(t)$ and $s_2(t)$ at time t.

Hint: use the values of $\mathbb{E}[p(\cdot)]$ and $\gamma_p(0)$ determined in your answers to Questions 1-b) and 1-c), respectively.

[7 Marks]

2. Consider the stochastic process $y(\cdot)$ generated by the state equations below:

$$\begin{cases} x_1(t+1) = \frac{1}{10} \cdot x_1(t) + \frac{10}{3} \cdot u(t) + \frac{7}{2} \cdot v(t) \\ x_2(t+1) = \frac{3}{5} \cdot x_1(t) + \frac{2}{5} \cdot x_2(t) + v(t) \\ y(t) = \frac{1}{2} \cdot x_2(t) \end{cases} \tag{2.1}$$

where $u(t)$ is a known exogenous input and $v(\cdot) \sim WN(0,1)$ is a white process.

a) Determine a suitable ARMAX model describing the stochastic process $y(\cdot)$

Hint: combine the above two state-space equations in (2.1) *to get an input-output stochastic model representation for* $y(\cdot)$.

[6 Marks]

b) Write the difference equation representing the optimal two-steps ahead predictor $\hat{y}(t|t-2)$ of $y(t)$ from past measured values of $y(\cdot)$.

[9 Marks]

c) Compute the variance of the prediction error $\varepsilon_2(t) = y(t) - \hat{y}(t|t-2)$.

[5 Marks]

3. Consider the following stochastic process:

$$y(t) = \frac{1}{3} \cdot \eta(t) + \frac{2}{3} \cdot \eta(t-2)$$

where $\eta(\cdot) \sim WN\left(0, \frac{1}{4}\right)$ is a white process.

a) Determine the correlation function $\gamma_y(\tau)$ for every possible value of τ.

[4 Marks]

b) Consider the following family of models

$$\mathscr{M}_1: \quad y(t) = \alpha \cdot y(t-2) + \xi(t), \qquad \xi(\cdot) \sim WN\left(0, \lambda^2\right)$$

Determine the optimal values of α and λ^2 according the *prediction error minimisation* (PEM) criterion.

[6 Marks]

c) Consider the following family of models

$$\mathscr{M}_2: \quad y(t) = \beta \cdot y(t-2) + \delta \cdot y(t-3) + e(t), \qquad e(\cdot) \sim WN\left(0, \mu^2\right)$$

Determine the optimal values of β, δ and μ^2 according the *prediction error minimisation* (PEM) criterion.

[6 Marks]

d) Evaluate the performance of models $\mathscr{M}_1$ and $\mathscr{M}_2$ determined in your answers to Questions 3-b) and 3-c) and compare the results. Discuss your findings.

[4 Marks]

4. A common way to model the kinematic parameters of a vehicle on a highway is described by the following differential equations:

$$\begin{aligned} \frac{d}{dt}v(t) &= a \cdot \left[1 - \left(\frac{v(t)}{v_{ref}}\right)^4 - \left(\frac{S}{d(t) - p(t)}\right)^2\right] \\ \frac{d}{dt}p(t) &= v(t) \end{aligned} \qquad (\star)$$

where

- $p(t)$ is the vehicle's position;
- $v(t)$ is the vehicle's velocity;
- $d(t)$ is the position of the next vehicle in the line.

Moreover, the model $(\star)$ is characterised by the following driver's parameters:

- v_{ref}: desired driver's velocity;
- a: vehicle's maximum acceleration;
- S: given parameter related to the safety gap that the driver want to maintain with respect of the next vehicle.

The vehicle is equipped with two ideal sensors that log the position $p(t)$ of the vehicle and the position $d(t)$ of the vehicle immediately ahead. The two sensors are synchronised and work with the same sampling time $h = 1\,\text{sec}$.

a) Provide a discrete-time description of the continuous-time model using the Euler discretisation method with the same sampling time of the two sensors.

Hint: the Euler method approximates the time-derivative with

$$\frac{d}{dt}x(k \cdot h) \simeq h^{-1} \cdot [x((k+1) \cdot h) - x(k \cdot h)].$$

where $k \cdot h$ and $(k+1) \cdot h$ denote two subsequent discrete-time instants and k is an integer index.

[4 Marks]

b) Rewrite the discrete-time model in the form

$$\rho_0(k) = \sum_{i=1}^{m} \vartheta_i \cdot \rho_i(k)$$

where k is the above discrete time index, ρ_i, $i = 0, \ldots, m$, are suitable functions computable using the data provided by the sensors, and ϑ_i, $i = 1, \ldots, m$ are suitable unknown parameters to be estimated.

[8 Marks]

c) Devise a least-square procedure to estimate the driver's parameters a, v_{ref} and S.

[8 Marks]